

```
import java.util.Scanner;

class QueueArray {

    private int[] arr;
    private int front, rear;
    private int capacity;

    public QueueArray(int capacity) {
        this.capacity = capacity;
        arr = new int[capacity];
        front = rear = 0; // Initially both pointers point to the beginning
    }

    public boolean enqueue(int element) {
        if (isFull()) {
            System.out.println("Queue is full!");
            return false;
        }
        arr[rear] = element;
        rear = (rear + 1) % capacity; // Circular queue implementation
        return true;
    }

    public int dequeue() {
        if (isEmpty()) {
            System.out.println("Queue is empty!");
            return -1; // Can modify to throw an exception
        }
        int element = arr[front];
        front = (front + 1) % capacity;
        return element;
    }

    public int peek() {
```

```
    if (isEmpty()) {  
        System.out.println("Queue is empty!");  
        return -1; // Can modify to throw an exception  
    }  
    return arr[front];  
}
```

```
public boolean isEmpty() {  
    return front == rear;  
}
```

```
public boolean isFull() {  
    return (rear + 1) % capacity == front;  
}
```

```
public void displayQueue() {  
    System.out.print("[");  
    for (int i = front; i != rear; i = (i + 1) % capacity) {  
        System.out.print(arr[i] + ", ");  
    }  
    System.out.print(arr[rear] + "]); // Include the last element  
}  
}
```

```
public class QueueDriver {
```

```
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);
```

```
        System.out.print("Enter queue capacity: ");  
        int capacity = scanner.nextInt();
```

```
        QueueArray queue = new QueueArray(capacity);
```

```
        int choice;
```

```
do {
    System.out.println("\nQueue Operations:");
    System.out.println("1. Enqueue");
    System.out.println("2. Dequeue");
    System.out.println("3. Peek");
    System.out.println("4. Display Queue");
    System.out.println("5. Exit");

    System.out.print("Enter your choice: ");
    choice = scanner.nextInt();

    switch (choice) {
        case 1:
            System.out.print("Enter element to enqueue: ");
            int element = scanner.nextInt();
            queue.enqueue(element);
            break;
        case 2:
            int dequeuedElement = queue.dequeue();
            if (dequeuedElement != -1) {
                System.out.println("Dequeued element: " + dequeuedElement);
            }
            break;
        case 3:
            int frontElement = queue.peek();
            if (frontElement != -1) {
                System.out.println("Front element: " + frontElement);
            }
            break;
        case 4:
            queue.displayQueue();
            break;
        case 5:
            break;
        default:
```

```
        System.out.println("Invalid choice!");
        break;
    }
} while (choice != 5);

scanner.close();
System.out.println("Program terminated.");
}
}
```